

Lab: Create TDD and BDD Test Cases



Estimated duration: **30 minutes**

Learning objectives

By the end of this lab, you will be able to:

- Write a TDD-style test objective based on business logic
- Create multiple BDD scenarios using Gherkin syntax
- Translate business requirements into developer- and tester-ready language
- Identify gaps in test coverage for common application features

Overview

In this lab, you will **create and document test cases** that align with **Test-Driven Development (TDD)** and **Behavior-Driven Development (BDD)** practices. While TDD focuses on designing system logic through tests, BDD emphasizes business behavior using plain-language scenarios.

You will apply concepts from the previous role-play lab, using a login feature as the business scenario. You'll write one developer-style TDD test idea and multiple Gherkin-based BDD scenarios to reflect how users interact with the system.

This lab reinforces your role in translating business rules into structured tests, bridging the gap between users and developers.

Key tasks

As part of this lab, you will:

- Define a TDD-aligned test case for a Login system
- Write BDD scenarios for multiple login outcomes
- Identify missing edge cases or risks in coverage
- Compile test documentation for reuse in real projects

Conceptual background: Login system

You are supporting the QA and development team for a **banking application**. The login system has the following business rules:

Requirement ID: REQ-LOGIN-01

"Users must be able to log in using their registered email and password. Unregistered users should see an error. After three failed attempts, the system should lock the account."

These rules form the basis for writing both technical and behavior-driven test cases.

Step-by-step walkthrough: Writing TDD and BDD test cases

Follow these steps to complete your analyst documentation:

Step 1: Review the business rules

- Registered users should log in with their email and password
- Unregistered users should see an error
- After three failed login attempts, the account should be locked

Step 2: Write a TDD-style test case idea

- Think like a developer: What would fail if the logic was missing?
- Describe the **input**, the **expected outcome**, and the **business rule** behind it

Step 3: Write multiple BDD scenarios

- Use Gherkin Given-When-Then syntax
- Include both positive and negative flows
- Use clear and consistent formatting

Step 4: Identify edge cases

- Think about errors, misuse, or unusual inputs
- Help the team avoid missed functionality or vulnerabilities

Your task: Create TDD and BDD test cases for REQ-LOGIN-01

Complete the following numbered instructions:

1. Write a TDD test objective

Write a developer-facing test objective that validates business logic for account logout. Example:

Prevent login after three failed attempts by locking the user account.

2. Write three BDD scenarios

Use Gherkin syntax to describe the following use cases:

- Successful login
- Login with an unregistered email
- Account logout after failed attempts

Scenario: Successful login with registered credentials

Given I have a registered account

And I enter my correct email and password

When I click the "Login" button

Then I should be taken to the dashboard

Scenario: Login with unregistered email

Given I do not have an account

And I enter an unregistered email and any password

When I click the "Login" button

Then I should see an error message: "User not found"

Scenario: Account logout after three failed login attempts

Given I have a registered account

And I enter an incorrect password three times

When I attempt to login again

Then my account should be locked

And I should see an error message: "Account locked. Please reset your password."

3. List three edge case considerations

Document risks or gaps that the team should also test:

- What if a user enters a valid email but leaves the password blank?
- What happens if the login page is refreshed after two failed attempts?
- Can bots abuse the login form for brute-force attacks?

4. Summarize analyst contributions

In 2–3 sentences, describe how your test cases help developers and testers build reliable login functionality.

Save and share your work

1. Save your TDD objective, BDD scenarios, edge case list, and analyst summary in a single file named:

create-tdd-bdd-cases.docx

2. Upload your file to the **course peer review area** for feedback.

Summary

In this lab, you learned how to:

- Write a TDD-style test case that validates business logic
- Use Gherkin syntax to describe realistic user interactions
- Translate business requirements into developer- and tester-friendly formats
- Think critically about edge cases and test coverage gaps
- Communicate test ideas clearly for better collaboration between QA and development teams

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